

Ch # 6

Video

⇒ Video is an excellent tool for delivering multimedia.

⇒ Video places the highest performance demand on computers and its memory and storage.

⇒ Video is the technology for electronic capture, storage, transmission, and reproduction of images and motion pictures.

⇒ A natural video scene is composed of multiple objects each with their own characteristics shape, depth, texture, and illumination.

⇒ Video information travels in the form of light waves and so must be converted into electrical signals using transducers like video cameras.

⇒ it is like a series of still images.

⇒ it uses RGB color space.

How video works?

- ⇒ 1. When light reflected from an object passes through a video camera lens.
- ⇒ 2. That light is converted into an electronic signal by a special sensor called a charged coupled device (CCD).
- 3. Top quality broadcast cameras and even camcorders may have as many as 3 CCD's (one for each hue of Red, Green, & Blue) to enhance the resolution of the camera.
- 4. There are several video standards for managing CCD output each dealing with the amount of separation between the components of signal.
- Q. The more the separation of the color information found in the signal the higher the quality of the image.

Q-What is Analog video?

- ⇒ Analog video is a video signal transferred by an analog signal.
- ⇒ When combined in to one channel, it is called Composite video.
- ⇒ Analog video may be carried in separate channels, as in two channels - video (YC) and multi-channel component video formats.

Q-What are Analog video transfer methods?

- o Composite video
- o S-video
- o Component video

Q-What is Composite video?

- ⇒ Composite video combines chrominance (color) and luminance (brightness) information into a single analog video transmission.
- ⇒ Color and intensity signals are mixed into a single carrier wave.
- ⇒ Chrominance is a composition of two colors (I and Q or U and V).

⇒ ⇒ The Composite signal yields less precise color definition, which cannot be manipulated.

⇒ it is used by broadcast TV, in NTSC TV e.g.

⇒ ⇒ The chrominance and luminance

1 Components can be separated at the receiver end and then the two color components can be further recombined.

⇒ When connecting to TVs or VCRs

Composite video uses only one wire and video color signals are mixed; not sent separately.

⇒ The audio and sync signals are additional to this one signal.

⇒ A composite video cable, often known as an RCA or "yellow plug" cable is

an outdated standard for transmitting video signal via a single cable or connector.

⇒ As video signal goes over a single wire, it is significantly compressed, resulting in loss of resolution and clarity.

⇒ it is also called SD video.

⇒ it comprises three types: NTSC, PAL, SECAM.

o it carries standard-definition video at 480i or 576i resolution.

History

o when color was introduced to black and white television in 1954, the composite video was born.

o The original monochrome signal (Y) was multiplexed with two color channels (U and V) and broadcast on the same TV channel.

⇒ it is also known as Radio Corporation of America (RCA) video.

⇒ Also called CVBS video. CVBS stands for Composite Video broadband signal or Color, Video, blanking, and Sync.

Advantages

o The main advantage of composite video is bandwidth is saved and fewer connection ports are required.

= What is Component video?

=> Component video transmission is a format of analog video transmission.
=> Component video divide video transmissions into three cables:

- =
 - o Red
 - o Green
 - o Blue

each of which transmit different part of the video signal.

=> The signal's brightness information is transmitted via the green cable.

The blue and red connections transfer the picture's blue and red components.

=> Component cables initially supplanted Composite as the default analog video cable due to their superior picture

quality and HD compatibility.

=> most video games consoles, DVD players, digital video recorders, cable set-top boxes, and Blu-ray players support Component video.

=> For any color separation scheme,

Component video gives the best color reproduction. it is preferred method for higher quality.

What is S-video?

=> S-video is also known as "Separate video" and is sometimes referred to as "Super video". Also called Y/C video.

=> This video analog transmission conveys information in two separate signals:

- o Chroma (color)
- o Luma (Brightness)

=> it only carries standard definition video and does not combine it with audio signals in a single wire.

=> There is less cross-talk between the color information and the crucial gray-scale information.

Q What are analog video display standards?

- o NTSC
- o PAL
- o SECAM

NTSC : - Video are recorded on the cassette.

=> Stands for National television standards Committee.

=> These standards define a method for encoding information into an electronic signal that creates a television picture.

=> It has a screen resolution of 625 horizontal scan lines and a scan rate of 25 frames per second.

=> Developed in North and Central America.

PAL -> Developed in Germany.

=> Stands for Phase alternate line.

=> PAL has screen resolution of 625 horizontal lines and a scan rate of 25 frames per second.

=> The refresh rate is 25 frames per second.

SECAM:

=> Stands for Sequential Color and memory.

=> SECAM has a screen resolution of 625 horizontal lines and is a color system.

=> Differs from the two because of its technology and broadcast method.

=> Developed in France and Russia, parts of Africa.

Q. What is digital video?

Definition:

=> Digital video is audiovisual content in a binary format, with information is presented as a sequence of digital data rather than in a continuous signal as analog information is.

=> Digital video comprises a series of orthogonal bitmap digital images displayed in rapid succession at a constant rate.

=> Digital video combines features of graphics and audio to create dynamic content for multimedia products.

=> The video and audio data are compressed before being written to a tape or digitally stored.

=> The video and audio data are compressed before being written to a tape or digitally stored.

=> Digital video is captured from television, VCR and camera sources.

=> Digital video solves generational loss.

=> Consist of a format for encoding and playing back video files by a computer.

= What is codecs?

=> Codecs are Compression technologies and have two Components:

Encoder: to Compress the files.

Decoder: to decompress the files.

=> Codecs can be Lossy and Lossless.

=> A codec is a Compiled program that uses Compression to shrink a large movie file

=> Codec stands for enCoder / DeCoder or Compressor / Decompressor

=> The purpose of codecs is to reduce the size of digital audio samples and video frames in order to speed up transmission and save storage space.

=> Because video and music files are large, they become difficult to transfer across the internet quickly.

=> Without codecs, downloads would take close to five times longer than they do now

Lossy codecs

- o MP3
- o Ogg Vorbis, AAC
- o m4a, mp4, audio etc

Lossless codecs

- o FLAC
- o ALAC
- o AIFF, Wave

=> Codecs are algorithms used to compress a video for delivery.

= Define MPEG?

=> MPEG stands for moving picture experts group

=> MPEG is a real-time video compression algorithm.

=> Browser support varies.

=> it is developed and maintained by the moving picture experts group, a working group of ISO and IEC.

=> MPEG uses a lossy form of compression.

=> MPEG-1, MPEG-2, MPEG-4, MPEG-7, MPEG-A

are several types of MPEG.

Advantage:

- o High Compression efficiency
- o widely supported
- o Good quality
- o Streamable
- o Scalable
- o Versatile.

Cons:

- o Lossy Compression
- o Limited color depth
- o Complexity

= What is playback?

- o Playback refers to the capability of playing recorded or pre-recorded video content.
- o It allows users to watch videos on various devices and involves functions like starting, stopping, pausing, rewinding, and fast forwarding.
- o Playback can be done through physical media, streaming services, or local storage devices.
- o Online streaming platforms gained popularity, offering on-demand playback with additional features like subtitles, looping, slow motion etc.

What is chroma key?

=> Also known as green screen or blue screen, is a technique used in video and multimedia production to superimpose one image or video into another based on color hues.

=> It involves replacing a specific color (typically green and blue) in a video with different background or images.

=> It is commonly used in filmmaking, television production, and multimedia projects.

- o It allows creators to place objects at any location or environment without physically being there.
- o Allow you to choose a color or range of colors that become transparent, allowing the video image to be seen through the computer image.

What is flickering?

- o Flickering in multimedia refers to the rapid and repetitive change in brightness or color intensity of visual elements.
- Flicker is visible fading between cycles displayed on video displays, especially the refresh interval on cathode ray tube based computer screens.

o It can be caused by mismatched refresh rates (the number of times in a second that display hardware draws the data), electrical issues, incompatible codecs or formats or it may be an intentional artistic choice. Troubleshooting options include checking display settings, updating drivers.

= Q. What is overscan & underscan?

=> Question and underscan are terms

= used in multimedia to describe how

an image is displayed on a screen in relation to the visible area.

= Question:

=> refers to enlarging the image beyond its actual boundaries of the screen.

=> it ensures that the entire image

fills the visible area of the screen and hiding any potential black borders or edges.

= Underscan:

=> refers to displaying an image that is smaller than the visible area of the screen.

=> Black borders or edges displayed around here.

< Q. What is safe title area?

=> It refers to a region within the

video or graphic frame where important visual elements should be placed to ensure

they are fully visible on different display devices.

=> It accounts for potential cropping or obscuring of the image due to overscan.

=> Older CRT televisions used to display less than the full image, so the same title area presented important content

from being cut off. While overscan is less prevalent

with modern displays, it is still a good practice to keep critical

information away from the frame's extreme edges to accommodate potential variations in overscan.

Q. What is clip?

=> A segment of video (or audio) recording.

=> It refers to short segment or portion of a video recording.

=> often interchangeably with "footage" or "sequence"

=> it represents a continuous section capturing a specific event or scene.

=> video clips are commonly used in

the post production process, where editors assemble and arrange various

clips to create a final video.

- Clips are pre-rendered into
ready that help create a seamless and
smooth video.

What is real time?

= when a video signal is broadcast
live on unedited.

What is frame?

=> A frame is a single still image
in a video.

=> it refers to a picture that makes
up a sequence of images displayed
rapidly to create the illusion of
motion.

=> video composed of many frames that are
displayed at specific rate.

=> Each frame contains visual information
and contributes to the overall video.

What is frame rate?

=> it refers to the number of frames or
images to be displayed per second in a
video.

=> it determines the smoothness of motion.
Common frame rates include 24 fps,
30 fps and 60 fps.

=> Higher frame rates result in smoother
motion but require more storage &

Processing power.

=> Lower frame rates may result in choppy
motion.

=> Choosing the right frame rate is essential
for achieving the desired visual quality and
effort in a video.

Q. What is streaming video?

=> Streaming video is a technique
refers to the real time delivery of
video content over the internet, allowing
users to watch videos without downloading
the entire file.

=> it is popular due to high-speed
internet availability.

=> Streaming relies on buffering and
continuous downloading of small portions
of the video.

=> Coders are used to encode and
decode the data.

=> Streaming provides benefits of
on demand viewing, but requires a
stable internet connection for un-
interrupted playback.

= Q. Size of video file depend on what factors?
o it is directly related to frame rate
= size and color-depth

Q. What are some video player software?

- o Apple QuickTime
- o Microsoft Windows media player
- o Adobe flash player

Q. What are video editing software?

- o Windows movie maker
- o Apple's iMovie
- o Apple final cut pro
- o Adobe Premiere
- o Batch processing

Q. What are video devices?

- o Video Camera
- o Video Capture card
- o Video Tuners

Q. What is refreshing?

o it is the transfer of data between two types of the same storage so there are no bitrate changes or alteration of data.

o Refresh rate refers to the number of times per second the content on a display screen is updated or redrawn. It is measured in Hertz (Hz). The

Standard refresh rate is 60 Hz.

Higher refresh rate means a high frame rate.

Q# Describe different video file formats?

(1) Mov (Movie):

- o Introduced by Apple in 1991.
- o Developed for Quicktime video player.
- o Also known as quick time movie format.
- o Best used for streaming.
- o High quality format that compressed video significantly without losing too much quality.

(2) WMV

- o Stands for windows media video.
- o Developed by Microsoft for its video player software.
- o Closely related to VMA.
- o Not played well by all systems.
- o Recent versions include more security measures.

(3) AVI

- o DRM = Prevent unauthorized duplication.
- o it is for streaming video.
- o Stands for Audio Video Interleave.
- o Introduced by Microsoft in the early 1990s for windows.

Discuss the principles of video compression technologies.

- o Not a streaming video format, so if it is played from a website, the entire file needs to be downloaded before viewing.
- o Can be played by most video players.
- o Because of the structure of the format, a player that supports some AVI cannot necessarily play all AVI files.
- o Also a strong internet presence.